

Keran QIAO

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EDUCATION

Shenzhen University

Shenzhen, Guangdong, China

Bachelor of Electrical Information Engineering (AI Elite Class)

Average Score: 85.6/100

Sep.2023-Present

- **Core Courses:** Information Theory and Coding Technology (94), Computer Vision Project Combat (94) Graph Deep Neural Networks (94), Intelligent Speech Processing (93), Deep Learning Framework Setup (93) Deep Reinforcement Learning (93), Introduction to Robotics (93), Industrial Machine Vision Technology (92)
- **Research Interest:** Parameter-Efficient Fine-Tuning, Large Language Models, Model Compression

RESEARCH EXPERIENCE

Academician-led Science and Innovation Training Camp

Academician Li Qingquan's Group: Ubiquitous Sensing and Spatial Intelligence

May.2024-Present

- Responsible for 3DGS model lightweighting and edge deployment on Jetson Orin NX. Implemented pruning strategies to remove redundant Gaussians with near-zero opacity or excessive scale, and optimized rendering pipelines to reduce memory footprint.

Provincial Innovation and Entrepreneurship Training Program for Undergraduate

Intelligent Ring Interaction and Authentication System for XR Devices

Jun.2025-May.2026

- Responsible for implementing a shared-parameter multi-task learning framework based on Progressive Layered Extraction (PLE) and an adaptive multi-scale convolutional expert network to jointly process bioimpedance and IMU signals, achieving accuracies of 97.8% for gesture recognition and 97.5% for user authentication while supporting lightweight edge deployment.

Independent Research Project

HiLA: Historical Layer Attention for Parameter-Efficient Fine-Tuning

Jun.2026-Present

- Research Focus: Parameter-efficient fine-tuning for large language models through historical layer-transition attention and LoRA-style constrained update routing; achieved 65.10 average accuracy on OpenCompass commonsense benchmarks, reaching state-of-the-art performance within the matched PEFT comparison. Manuscript in preparation for submission to ICLR 2027.

PATENT

[1] **Invention patent:** "A Causal Reasoning-Based Method and System for Multimodal Model Compression and Distillation", A method for compressing multi-modal models using causal inference to guide pruning, quantization, and distillation while preserving critical causal relationships

First Student Author, Authorized, Patent No. CN202511804208.3

Feb.2026

WORK EXPERIENCE

Research Assistant

Sep.2024-Jan.2026

State Key Laboratory of Radio Frequency Heterogeneous Integration, Shenzhen University

- Conducted systematic benchmarking and parametric trade-off analysis on RF heterogeneous integration equipment, evaluating process capabilities, system compatibility, and performance constraints to establish optimal integration architecture for lab-scale prototyping. Translated technical findings into strategic equipment selection and process configuration recommendations, directly informing R&D decision-making and integration roadmaps.

Research Assistant

Sep.2024-Present

Guangdong Provincial Engineering Center for Multimedia Information Services

- Conducted systematic research on Large Language Model compression and parameter-efficient fine-tuning,

encompassing experimental investigation and performance analysis of pruning, quantization, and distillation techniques, as well as the design and evaluation of PEFT frameworks based on low-rank adaptation.

AWARDS

National Level & Above

- **Global First Prize** **Huawei ICT Competition (Ascend AI Track)** **Apr.2026**
Rank: **1st** (Team Leader) – End-to-end LLM development & deployment
- **National First Prize** **China Robot and Artificial Intelligence Competition** **Jul.2026**
Rank: **1st** (Team Leader) – Frontend, backend & agent algorithm development
- **National First Prize** **China Robot and Artificial Intelligence Competition** **Jul.2026**
Rank: **2nd** (Core Member) – Edge AI algorithm development
- **National First Prize** **National College Student IoT Design Competition** **Aug.2025**
Rank: **2nd** (Core Member) – Edge AI algorithm development
- **National Third Prize** **iCAN Innovation Competition (AI Challenge)** **Dec.2025**
Rank: **2nd** (Core Member) – Edge AI algorithm development

Provincial Level

- **Provincial First Prize** **Huawei ICT Competition (Network Track)** **Dec.2024**
Rank: **Individual Competitor**
- **Provincial First Prize** **"Siemens Cup" China Intelligent Manufacturing Challenge** **Jul.2026**
Rank: **2nd** (Core Member) – Edge AI algorithm development
- **Provincial First Prize** **National College Student Mathematical Modeling Competition** **Dec.2024**
Rank: **2nd** (Core Member) – Mathematical modeling
- **Provincial Second Prize** **National College Student Mathematical Modeling Competition** **Dec.2025**
Rank: **3rd** (Core Member) – Mathematical modeling
- **Provincial Second Prize** **"Datang Cup" 5G Technology Competition** **Apr.2025**
Rank: **1st** (Lead Contributor) – 5G communication system simulation
- **Provincial Second Prize** **iCAN Innovation Competition** **Oct.2025**
Rank: **2nd** (Core Member) – Edge AI algorithm development

SCHOLARSHIP

- **Star of SZU (Top 1%), Shenzhen University** **2025-2026**
- **Excellent Student Leader (Second Prize), Shenzhen University** **2025-2026**
- **Star of Double Innovations (Second Prize), Shenzhen University** **2025-2026**
- **Star of Double Innovations (Group) (First Prize), Shenzhen University** **2025-2026**
- **Star of Double Innovations (Group) (Second Prize), Shenzhen University** **2025-2026**

EXTRACURRICULAR ACTIVITY

Volunteer Work

- Contributed 600+ cumulative service hours across high-profile events, including the 15th National Games and the Shenzhen Marathon

Minister, College Volunteer Association (2024–2025)

- Spearheaded the planning and execution of multiple large-scale volunteer initiatives, overseeing team operations and volunteer training for the college.

SKILLS

Programming: Python, MATLAB, C++, LaTeX, Git

Platforms: NVIDIA Jetson Orin NX, Huawei Ascend AI, Raspberry Pi, Orange Pi

Languages: Chinese (Native), English (IELTS: 6.5)